



Technical Note

A new heat transfer correlation for supercritical water flowing in vertical tubes



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ARTICLE INFO

Article history:

Received 15 November 2013

Received in revised form 23 March 2014

Accepted 19 June 2014

Available online 17 July 2014

Keywords:

Supercritical pressure

Water

Heat transfer

Correlation

ABSTRACT

Supercritical water has been used in many industrial fields, such as fossil-fired power plants and nuclear reactors, where the determination of heat transfer coefficients is required. Although many empirical correlations for heat transfer coefficients of supercritical water have been proposed, their prediction accuracy is not satisfactory, and thus more accurate correlations are needed. This paper proposes a new correlation for heat transfer of supercritical water flowing in vertical tubes based on 5366 experimental data points obtained from 13 independent papers. It has a mean absolute deviation (MAD) of 5.4% and predicts 95.7% of the entire database within $\pm 15\%$, while the best existing correlation only has an MAD of 13.6% and predicts 64.4% of the database within $\pm 15\%$, which demonstrates that the new correlation is much better than any existing one.

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1. Introduction

Due to the unique properties and favorable heat and mass transfer characteristics, supercritical water has attracted great interest in many applications, such as fossil-fired power plants and nuclear reactors. Supercritical water-cooled nuclear reactors (SCWRs) are high pressure (about 25 MPa) and high temperature (up to 625 °C) reactors that operate above the thermodynamic critical point of water (22.064 MPa and 373.95 °C), offering potential for high thermal efficiencies, considerable plant simplifications, and better safety and economy [1]. Under supercritical conditions, the thermophysical properties of water vary dramatically, causing deviant heat transfer characteristics compared to that of subcritical fluids, and thus the heat transfer of supercritical water cannot be fairly predicted with conventional heat transfer correlations [2]. Therefore, the study of heat transfer correlations for supercritical water is necessary.

The investigations of heat transfer of supercritical water have been carried out since the 1930s. Many researchers reviewed and summarized the existing experimental and theoretical studies. Since the heat transfer mechanisms occurring in the supercritical fluids have not been distinctly understood, the prediction of heat transfer coefficients for supercritical fluids is mainly conducted using empirical approaches, and many correlations were proposed in the open literatures based on experimental data [3]. In order to

evaluate which correlations could provide more accurate predictions for heat transfer of supercritical water, some investigators compared existing correlations with different experimental data.

Licht et al. [4] conducted an experimental investigation of supercritical water heat transfer in a circular and a square annular flow channel, with the parameter range of mass flux of 350–1425 kg/m² s, heat flux up to 1.0 MW/m², and bulk inlet temperature up to 400 °C at a pressure of 25 MPa. They compared the correlations of Dittus and Boelter [6], Krasnoshchekov et al. [7], Watts and Chou [8], and Jackson [5] with the experimental data and found that the Jackson [5] correlation was able to predict the test data best, capturing 86% of the data within 25%, and that the Watts and Chou [8] correlation showed a similar trend but underestimated the measurements by 10% relative to the Jackson [5] correlation.

Yu et al. [9] compiled 1142 experimental data points of supercritical water heat transfer with mass flux of 90–2441 kg/m² s, heat flux of 90–1800 kW/m², tube hydraulic diameter of 1.5–38.1 mm, and pressure of 22.6–41.0 MPa, with which 17 heat transfer correlations were assessed. The results showed that the Bishop et al. [10] correlation performed best. Yu et al. [9] developed a new correlation and stated that it is a little better than the Bishop et al. [10] correlation.

Zhu et al. [11] conducted an experimental investigation of supercritical water heat transfer in a tube with an inner diameter of 26 mm, with the parameter range of pressure up to 30 MPa, mass flux from 600 to 1200 kg/m² s, and heat flux from 200 to 600 kW/m². With their experimental data, they evaluated the

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Nomenclature

c_p	specific heat at constant pressure (J/kg K)
$\bar{c}_p$	average specific heat (J/kg K), $(h_w - h_b)/(t_w - t_b)$
D	inner tube diameter (m)
G	mass flux (kg/m ² s)
Gr	Grashof number $(gD^3(\rho_b - \rho_w)/\rho\nu^2)$
Gr^*	Grashof number based on heat flux $(g\beta D^4 q/\lambda\nu^2)$
g	acceleration due to gravity (m/s ²)
L	tube length (m)
Nu	Nusselt number $(\alpha D/\lambda)$
$\overline{Pr}$	average Prandtl number $(\mu\bar{c}_p/\lambda)$
p	pressure (Pa)
q	heat flux (W/m ²)
Re	Reynolds number (GD/μ)
T	temperature (K)
t	temperature (°C)

Greek symbols

α	heat transfer coefficient (W/m ² K)
β	thermal expansion coefficient (1/K)
λ	thermal conductivity (W/m K)
μ	dynamic viscosity (Pa s)
ν	kinematic viscosity (m ² /s)
ρ	density (kg/m ³)

Subscripts

b	at bulk temperature
exp	experimental
in	inlet
pc	at pseudo-critical temperature
$pred$	predicted
w	at inner wall temperature

correlations of Shitsman [12], Swenson et al. [13], Krasnoshchekov et al. [7], Jackson [5], and their own. Results showed that their own correlation and the Swenson et al. [13] correlation were the best according to the calculation results.

Mokry et al. [1] proposed a new correlation of supercritical water heat transfer based on their experimental data with the parameter range of mass flux from 200 to 1500 kg/m² s, heat flux up to 1250 kW/m², and inlet temperature from 320 to 350 °C at a pressure of 24 MPa, and compared it with the correlations of Dittus and Boelter [6], Bishop et al. [10], Swenson et al. [13], and Jackson [5]. They found that all the correlations deviated substantially from the experimental data within the pseudo-critical range except their own.

Jäger et al. [14] compiled a database of supercritical water heat transfer from six experimental data sources. The parameter range of the data cover mass flux of 500–2150 kg/m² s, heat flux of 116–1577 kW/m², tube hydraulic diameter of 7.5–26 mm, and pressure of 22.6–31.0 MPa. Based on the database, they assessed 15 correlations of supercritical water heat transfer. Results showed that none of them gave satisfactory predictions for all cases, especially in the pseudo-critical region, and that the Bishop et al. [10] correlation performed best.

The above review shows that the conclusions regarding the top heat transfer correlations for supercritical water are not consistent, and that more accurate correlations need to be developed. This work conducts a comprehensive survey of the correlations and experimental investigations of supercritical water heat transfer. Based on the experimental data compiled, the existing correlations are evaluated, and a new correlation is developed. Compared with the previous best counterparts, the new correlation shows far higher prediction accuracy.

2. Experimental data description

A comprehensive survey of experimental investigations of heat transfer to supercritical water shows that the majority of experimental data were obtained from vertical circular tubes [15]. There are 5366 data points compiled from 13 open papers as listed in Table 1. All the data were taken from vertical circular tubes and were presented graphically in the source papers. The commercial software GetData Graph Digitizer is used to translate the experimental data points on the figures into digital data. The NIST REFPROP software is used to determine the fluid thermophysical properties corresponding to the given experimental conditions.

The parameter ranges cover bulk enthalpy from 278 to 3169 kJ/kg, mass flux from 201 to 2500 kg/m² s, heat flux from 129 to 1735 kW/m², pressure from 22 to 34.3 MPa, and tube hydraulic diameter from 6 to 26 mm.

In general, the experiments confirmed that different flow and operating conditions resulted in various heat transfer characteristics which basically characterized as so-called “normal”, “deteriorated” and “improved” heat transfer regimes [16]. It is very difficult to provide the exact definitions of these heat transfer regimes. Nevertheless, many criteria were mentioned in the open literature [2], and a frequently-used one was proposed by Koshizuka et al. [26] as the following:

$$R = \alpha_{exp}/\alpha_0 \quad (1)$$

where R is the ratio of the heat transfer coefficients of the measured value α_{exp} to the reference value α_0 , which is calculated with the Dittus and Boelter [6] correlation that was developed for conventional single-phase pipe flow:

$$Nu_0 = 0.023 Re_b^{0.8} Pr_b^{0.4} \quad (2)$$

The heat transfer of supercritical water flow is deemed to be normal when $0.3 \leq R \leq 1$, deteriorated when $R < 0.3$, and improved when $R > 1$. Based on this criterion, the 5366 experimental data points are divided into three regimes as shown in Fig. 1, from which it can be seen that 3430 points (63.9%) are in the normal heat transfer regime, 1267 points (23.6%) in the improved heat transfer regime, and 669 points (12.5%) in the deteriorated heat transfer regime. As can be seen in Fig. 1, a peak in heat transfer coefficients near the critical and pseudo-critical points was recorded and most of the improved and deteriorated data points appeared near the pseudo-critical points.

3. Development of a new correlation

In order to develop a better correlation, the existing heat transfer correlations of supercritical water are analyzed and evaluated first using the 5366 experimental data. There are 20 correlations assessed, including those of Mokry et al. [1], Jackson [5], Krasnoshchekov et al. [7], Watts and Chou [8], Yu et al. [9], Bishop et al. [10], Zhu et al. [11], Swenson et al. [13], Yamagata et al. [18], Griem [20], Xu et al. [21], Petukhov et al. [27], Liu and Kuang [28], Gupta et al. [29], Kuang et al. [30], Ornatsky et al. [31], Gorbun' et al. [32], and Krasnoshchekov and Protopopov [33]. There are eight correlations having a mean absolute deviation (MAD) Less than 20%, as

Table 1
Experimental data sources of vertical circular tubes.

Reference	Flow range: $t(^{\circ}\text{C})/p_{\text{in}}(\text{MPa})/G(\text{kg}/\text{m}^2\text{ s})/q(\text{kW}/\text{m}^2)$	Geometry range: $D(\text{mm})/L(\text{mm})/\text{Material}$	Data points
Zhu et al. [11]	282–440(t_b)/23–30/600–1200/200–600	26/1000/1Gr18Ni9Ti stainless steel	120
Shitsman [12]	100–250(t_{in})/24.5–34.3/350–600/270–700	8,16/800,1600,3200/1Gr18Ni9Ti steel	331
Swenson et al. [13]	75–576(t_b)/23–41/542–2150/200–1800	9.42/1830/AISI-304 stainless steel	159
Mokry et al. [16]	320–350(t_{in})/24/200–1500/0–884	10/4000/12Cr18Ni10Ti stainless steel	1323
Vikhrev et al. [17]	50–425(t_b)/26.5/500–1900/230–1250	20.4/6000/1Gr18Ni10Ti stainless steel	424
Yamagata et al. [18]	230–540(t_b)/22.6–29.4/310–1830/116–930	7.5,10/1500,2000/AISI-316 stainless steel	250
Alekseev et al. [19]	100–350(t_{in})/24.5/380–820/100–900	10.4/750/Kh18N10T steel	163
Griem [20]	343–421(t_b)/22–27/300–2500/200–700	14/unmentioned/unmentioned	259
Xu et al. [21]	230–450(t_b)/23–30/600–1200/100–600	12/2800/1Gr18Ni9Ti stainless steel	260
Li et al. [22]	264–323(t_{in})/23–26/440–1521/189–1338	7.6/2640/Inconel-625 stainless steel	989
Pan et al. [23]	330–550(t_b)/22.5–30/1009–1626/216–822	17/2000/1Gr18Ni9Ti stainless steel	231
Wang et al. [24]	300–400(t_b)/23–26/450–1200/200–1200	10/2500/Inconel-625 stainless steel	688
Li et al. [25]	300–500(t_b)/23–25/600–1200/400–1100	6/3000/Inconel-625 stainless steel	169

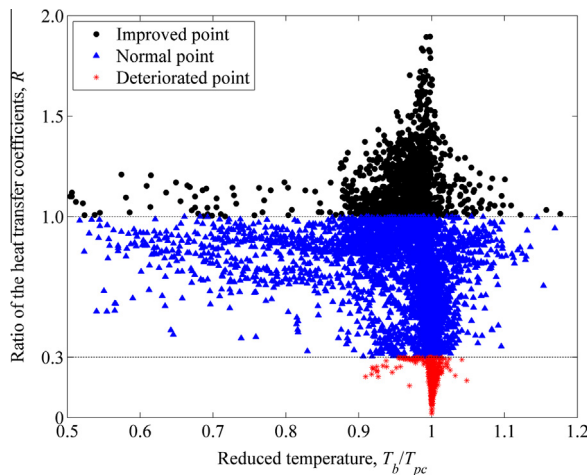


Fig. 1. Partition of total experimental data into three heat transfer regimes.

presented in Table 2, where MRD is the mean relative deviation and R_{15} is the percentile of the data points having the relative deviation (RD) within $\pm 15\%$ band. The MAD determines the average level of the prediction accuracy, and the MRD indicates the degree of the over-prediction or under-prediction.

$$\text{RD} = (\alpha_{\text{pred}} - \alpha_{\text{exp}}) / \alpha_{\text{exp}} \quad (3)$$

$$\text{MRD} = \frac{1}{N} \sum_{i=1}^N \text{RD}_i \quad (4)$$

$$\text{MAD} = \frac{1}{N} \sum_{i=1}^N |\text{RD}_i| \quad (5)$$

Based on the analysis of the existing correlations, a general form of heat transfer correlation for supercritical water heat transfer is proposed as the following:

$$\text{Nu}_b = c_0 X_1^{c_1} X_2^{c_2} \cdots X_k^{c_k} \quad (6)$$

where $X_1, X_2, \dots, X_k$ are the selected dimensionless groups, and $c_0, c_1, c_2, \dots, c_k$ are the constants needed to be determined. Taking the natural logarithm on the both sides of the above equation, it follows

$$\ln \text{Nu}_b = \ln c_0 + c_1 \ln X_1 + c_2 \ln X_2 + \cdots c_k \ln X_k \quad (7)$$

Eq. (7) is a typical multivariate linear regression model, whose least square solution can be given as

$$C = (X^T X)^{-1} X^T Y \quad (8)$$

where X , Y , and C are the matrices given by

$$X = \begin{bmatrix} 1 & \ln X_1(1) & \ln X_2(1) & \cdots & \ln X_k(1) \\ 1 & \ln X_1(2) & \ln X_2(2) & \cdots & \ln X_k(2) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & \ln X_1(n) & \ln X_2(n) & \cdots & \ln X_k(n) \end{bmatrix}_{n \times (k+1)} \quad Y = \begin{bmatrix} \ln \text{Nu}_b(1) \\ \ln \text{Nu}_b(2) \\ \vdots \\ \ln \text{Nu}_b(n) \end{bmatrix}_{n \times 1} \quad C = \begin{bmatrix} \ln c_0 \\ c_1 \\ \vdots \\ c_k \end{bmatrix}_{(k+1) \times 1} \quad (9)$$

where k and n are the number of the selected dimensionless groups and the sample data points, respectively. With the methods used by Fang and Xu [34] and Fang et al. [35], ten dimensionless groups from the best existing correlations are initially chosen. Then, through variable transformations the dimensionless groups are reduced to eight ($k=8$). Substituting the eight dimensionless groups into Eq. (6), the general form of heat transfer correlation for supercritical water heat transfer becomes

$$\text{Nu}_b = c_0 \text{Re}_b^{c_1} \text{Pr}_b^{c_2} (\bar{c}_p / c_{p,b})^{c_3} (\text{Pr}_w / \text{Pr}_b)^{c_4} (\nu_w / \nu_b)^{c_5} \text{Gr}_b^{c_6} (\text{Gr}_b^*)^{c_7} (q_b^+)^{c_8} \quad (10)$$

Based on the database of the 5366 experimental data points and the least square method, extensive computer tests are conducted to

Table 2
Deviations of correlations against the experimental data (%).

Correlations	Total data			Normal data			Improved data			Deteriorated data		
	MAD	MRD	R_{15}	MAD	MRD	R_{15}	MAD	MRD	R_{15}	MAD	MRD	R_{15}
New correlation	5.4	-0.7	95.7	5.5	-0.5	95.3	5.0	-1.8	96.7	5.6	0.3	95.7
Mokry et al. [1]	13.6	-2.4	64.4	11.6	-3.5	70.7	16.2	-8.8	51.9	18.9	15.4	55.5
Swenson et al. [13]	15.0	1.7	60.8	13.7	0.7	64.8	17.3	-0.3	51.4	17.2	10.9	57.8
Petukhov et al. [27]	15.7	6.5	61.5	11.7	6.0	72.9	14.2	-9.3	56.9	39.1	39.1	11.5
Liu-Kuang [28]	15.9	8.0	60.5	13.5	6.3	67.6	17.9	4.7	50.2	24.5	23.0	43.5
Gupta et al. [29]	17.5	-9.1	46.2	16.3	-11.4	46.7	21.2	-12.4	36.5	16.1	8.9	62.2
Watts-Chou [8]	18.2	8.0	55.7	13.9	7.0	65.0	14.6	-9.5	56.3	46.7	46.7	7.2
Zhu et al. [11]	18.8	14.2	56.7	15.4	12.3	63.8	16.6	5.7	59.3	40.2	39.9	15.5
Kuang et al. [30]	18.9	-0.4	49.3	15.9	-2.6	54.5	20.1	-11.2	42.3	32.2	31.1	35.9

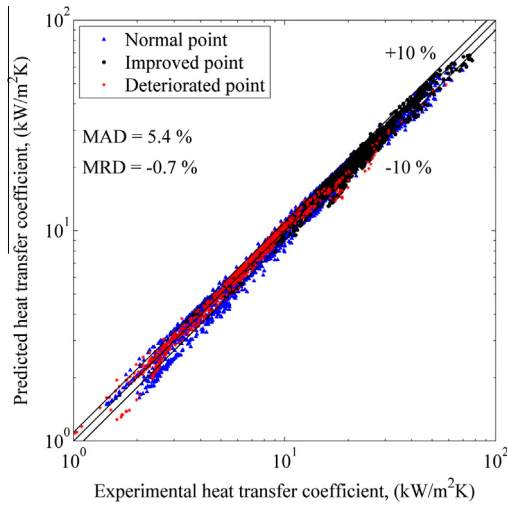


Fig. 2. Comparison of the new correlation with the entire experimental data.

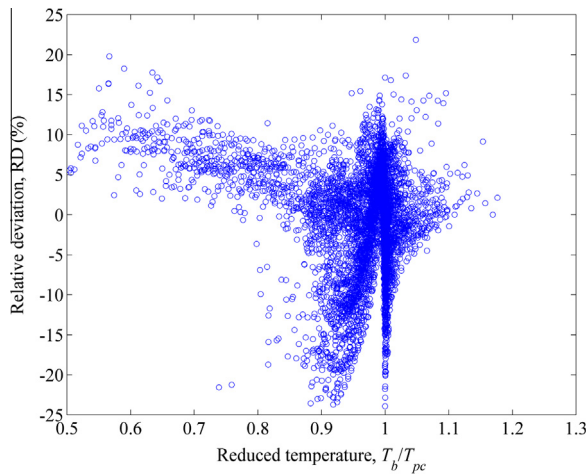


Fig. 3. The distribution of the relative deviations.

reduce the dimensionless groups and determine the regression constants in Eq. (10). The best form is found to be

$$Nu_b = 0.46 Re_b^{0.16} \left(\frac{Pr_w}{Pr_b} \right)^{0.1} \left(\frac{v_w}{v_b} \right)^{-0.55} \left(\frac{\bar{C}_p}{C_{p,b}} \right)^{0.88} \left(\frac{Gr_b^*}{Gr_b} \right)^{0.81} \quad (11)$$

The new correlation is compared with most cited existing correlations based on the database of the 5366 experimental data points. The result is listed in Table 2, from which it can be seen that the new correlation has an MAD of 5.4%, predicting 95.7% of the entire database within $\pm 15\%$, while the best existing correlation, the Mokry et al. [1] correlation, only has the MAD of 13.6% and predicted 64.4% of the entire database within $\pm 15\%$. In the normal, improved, and deteriorated heat transfer regimes, the best predictions of the existing correlations are the MAD of 11.6% (Mokry et al. [1] correlation), 14.2% (Petukhov et al. [27] correlation), and 16.1% (Gupta et al. [29] correlation) respectively, which shows that none of the existing correlations can be the best fit in all the three heat transfer regimes. However, the new correlation shows good performances, with the MAD of 5.5%, 5.0%, and 5.6% in the normal, improved, and deteriorated heat transfer regimes respectively. The above comparison results show that the new correlation improved the predication performance remarkably.

Fig. 2 shows the comparison of the predictions of the new correlation with the experimental database. It can be seen that the predictions agree with the measured data very well. Fig. 3 shows the RD distribution of the predictions of the new correlation with reduced temperature. It can be seen that small deviations appear outside the pseudo-critical region, while most of the big deviations occur in the vicinity of the critical and pseudo-critical points, where the thermophysical properties of supercritical water change rapidly with temperature and pressure.

Conclusions

- (1) The database containing 5366 experimental data points of heat transfer to supercritical water flowing in vertical tubes is compiled from 13 independent sources, covering the parameter range of bulk enthalpy from 278 to 3169 kJ/kg, mass flux from 201 to 2500 kg/m² s, heat flux from 129 to 1735 kW/m², pressure from 22 to 34.3 MPa, and tube hydraulic diameter from 6 to 26 mm. The entire database is divided into three regimes with 3430 points (63.9%) in the normal heat transfer regime, 1267 points (23.6%) in the improved heat transfer regime, and 669 points (12.5%) in the deteriorated heat transfer regime.
- (2) There are 20 existing correlations for supercritical water flowing in pipes are analyzed and evaluated with the database. The Mokry et al. [1] correlation performs best, with the MAD of 13.6% for the entire database. For all the studied correlations, the best predictions have the MAD of 11.6% for the normal regime, 14.2% for the improved regime, and 16.1% for the deteriorated heat transfer regime, none of which can be the best fit in all the three regimes.
- (3) A new correlation for supercritical water flowing in vertical tubes is developed based on the database. It has an MAD of 5.4% and predicts 95.7% of the entire database within $\pm 15\%$, while the best existing correlation only has an MAD of 13.6% and predicts 64.4% of the entire database within $\pm 15\%$. Besides, the new correlation predicts the normal, improved, and deteriorated heat transfer regimes with an MAD of 5.5%, 5.0%, and 5.6%, respectively, far better than any existing correlation for the given regime.

Conflict of interest

None declared.

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